

Building temperature regulation in a multi-zone HVAC system using distributed adaptive control^{*}

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ABSTRACT

During recent years there have been considerable research efforts on improving energy efficiency of buildings. Since Heating, Ventilation and Air-Conditioning (HVAC) systems are responsible for a big part of energy consumption, developing efficient HVAC control systems is crucial. In most of the developed approaches, precise knowledge of system parameters and/or adequate historical data is required. However, these approaches may not perform as well in the presence of dynamic parameter changes due to human activity, material degradation, and wear and tear, or disturbances and other operational uncertainties due to occupancy, solar gains, electrical equipment, and weather conditions. In this paper, we consider buildings with several climate zones and propose a distributed adaptive control scheme for a multi-zone HVAC system which can effectively regulate zone temperature by applying on-line learning and assuming exchange of information between neighboring zones. The controller of each zone achieves the local objective of controlling zone temperature by compensating for the effects of neighboring zones as well as for possible changes in the parameters of the system. Despite the exchange of information, each local controller does not know how the control actions and temperature of a neighboring zone affect the temperature of its own zone. For this reason, each local controller is estimating the parameters of the interconnections in real time and uses them together with the exchanged information to provide a more accurate local zone temperature control. The proposed method is illustrated using an example of temperature control in a six-zone building as well as a large school building, which are implemented in a Building Controls Virtual Test Bed (BCVTB) environment using EnergyPlus and MATLAB/Simulink.

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1. Introduction

Research has shown that buildings are one of the major energy consumers reaching up to 40% of final energy consumption in developed countries. Heating, Ventilation and Air-Conditioning systems (HVAC) are the largest contributors in building energy consumption, since they are responsible for 50% of final building energy needs [1]. Such systems are important for improving the quality of life in buildings, since they are necessary for providing comfortable temperature and air quality indoor conditions. Enhancing their efficiency can lead to reduced energy needs as well as improved climate conditions for occupants, while it can be a more cost-effective solution than replacing equipment with more modern technology [2]. Recent advances in sensors, electronics, and

communications have motivated a lot of research for designing and optimizing HVAC control systems for buildings, often referred to as smart buildings.

Developing appropriate mechanisms for operating and controlling HVAC systems is a challenging task for engineers and there have been several approaches on the subject. An accurate model of the building is crucial for the design of sophisticated HVAC systems. The zone model represents the heat dynamics of a space, the climate of which is controlled by an air-handling unit (AHU) or an air-terminal device [3,4]. Such a space is referred to as a thermal zone [5]. Air units are used to add or remove heat in order to modify zone temperature and overcome internal and external heat gains and losses. Although a building is usually divided into several zones, there exist centralized approaches that consider the whole building as an individual system. However, such schemes suffer from the disadvantages of centralized schemes, such as higher susceptibility to faults that affect temperature control, and higher computational demands. On the other hand, dividing a large building into multiple thermal zones and designing controllers for each zone leads to decentralized control schemes. Such control schemes

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are more robust to faults, since, if one zone fails locally, the fault does not affect the performance of the other zones. However, treating each zone as an isolated subsystem by ignoring the effects of neighboring zones may negatively affect performance. Strong interconnections may affect system performance if their impact cannot be attenuated, as analyzed in [6,7]. Air flow due to convection between thermal zones through internal openings is significant and can have a direct impact on zone temperature, since it results in heat transfer between zones, as explained in [8–10]. Neighboring zones also affect each other indirectly due to heat transfer by conduction through separating surfaces, as shown in [11] and in [12], where a feedback linearization scheme that reduces the heat transfer impact due to conduction is proposed. Therefore, a distributed approach that assigns to each thermal zone a local controller which takes into account the climate conditions of neighboring zones can combine tolerance to faults, which is a characteristic of a decentralized approach, with the high performance of a centralized scheme.

Various research efforts have suggested a plethora of different approaches in order to overcome the several challenges of designing a satisfactory HVAC control mechanism. One of the principal approaches is the use of Model Predictive Control (MPC) techniques. Model predictive control consists of optimizing a cost function that expresses system performance based on some metric, while satisfying constraints on the feasible control actions. A comprehensive report of the characteristics and advantages of MPC can be found in [13], where distributed approaches with zone climate controllers that consider the neighboring zones effect, such as in [14], are preferred. In [15] the authors used sequential quadratic programming and dual decomposition to develop a distributed MPC scheme, while in [16] the authors designed a distributed MPC control scheme based on Benders' decomposition. The authors in [17] present a two-level hierarchical MPC scheme. More MPC approaches are presented in [18–26]. Other applied control techniques include classical control and advanced nonlinear and fuzzy control [27], such as the application of a fuzzy cognitive map. A novel approach that combines optimization techniques and simulation has been introduced in [28], in order to create a "plug and play" scheme. Optimal control strategies have been proposed [29], as well as LQ and cascade control [30,31], optimization [32,33], data-driven [34–38] and simulation approaches [39]. Such schemes have the advantage to be robust with respect to disturbances or parameter changes, but most of them require accurate knowledge of system parameters.

Choosing the appropriate structure and strategy for HVAC control is a challenge that engineers are faced with. Many control approaches require knowledge of system parameters and dynamics, such as surfaces and thermal capacity of materials. However, dynamics change due to several factors, such as human activity or wear and tear. For example, the opening and closing of doors and windows changes drastically the surfaces that separate thermal zones and open doors create internal openings which lead to exchange of heat between zones due to convection. In addition, leakage in insulation may result in erroneous measurement of thermal capacity. Thus, a control scheme that needs exact and correct measurement of such parameters may prove to be problematic. Besides, there is often the case that surface temperature is uncontrolled, although it affects air temperature, which, in addition to disturbances such as outside weather conditions, heat gains from light or indoor human activity, can deteriorate system efficiency. The impact of the aforementioned uncertainties has been examined in [40].

Adaptive control does not require knowledge of system parameters, as it uses a parameter estimator in real time to estimate the unknown parameters and update the controller parameters accordingly. In [41] the authors proposed an adaptive and robust control

scheme for nonlinear HVAC systems, which provides a solution in the case of one thermal zone. However, large buildings are usually divided into multiple zones. In this direction, several adaptive control schemes have been proposed including decentralized schemes, such as in [42] where the authors consider zones that are enclosed in other zones and in [43] where interactions between zones are considered to be bounded disturbances, and, thus, their impact is ignored. In addition, centralized approaches include adaptive feed-forward control schemes, such as in [44], and adaptive event-triggered control, as proposed in [45]. In [46] the authors use machine learning and propose an adaptive hybrid meta-heuristic. Adaptive and learning-based approaches have also been proposed in [47,48]. Based on adaptive control and learning techniques for HVAC systems, combining the benefits of distributed control and at the same time removing the need for precise and up-to-date information on system parameters is of high practical importance.

In this paper, we focus on controlling an HVAC system with multiple zones using a distributed adaptive control approach which assumes exchange of state information between zones but does not require accurate knowledge of system parameters. First, we examine the ideal case when the control designer has exact knowledge of system dynamics and parameters and zone temperature, surrounding surfaces temperature, supply air temperature, and outside air temperature are available for measurement for each zone. Given these assumptions, we show how a distributed control scheme can effectively achieve satisfactory performance and temperature regulation, even when unknown heat gains from human activity or other sources are included in the system. The consideration of the ideal case indicates what is achievable in a perfect world where all parameters are known exactly and forms the reference for comparing with, under non-ideal conditions. In the case where the system parameters are unknown we use the same controller structure as in the ideal case and replace all the unknown parameters with their on-line estimates generated by an adaptive law or parameter estimator. The proposed distributed adaptive control scheme guarantees performance and robustness with respect to all aforementioned uncertainties without requiring knowledge of system parameters. We demonstrate the performance of the proposed scheme by applying it to a two-story building as well as to a large school building for temperature regulation. The buildings are modeled using EnergyPlus software and the control scheme is implemented in MATLAB/Simulink. The overall system is co-simulated using Building Controls Virtual Test Bed (BCVTB) environment. Simulations under several scenarios and different conditions have been conducted in order to show the performance of the proposed scheme.

The paper is organized as follows: Section 2 includes the notation and nomenclature regarding HVAC systems, while Section 3 presents the model of a multi-zone HVAC system and the control objectives. Section 4 presents the proposed control scheme and is divided into two subsections. The analysis for the case when the system parameters are known exactly is presented in Section 4.1. Section 4.2 presents the distributed adaptive control scheme for the case of unknown system parameters. Two case studies that demonstrate the application of the proposed control scheme to a two-story building and a large school building are presented in Section 5. Finally, the conclusion is presented in Section 6, while the Appendix includes the proofs of stability and robustness of the proposed control scheme.

2. Notation and nomenclature

The field of real numbers is denoted by $\mathbb{R}$. For any vector $x \in \mathbb{R}^n$, $\|x(k)\|$ represents the Euclidean (l_2) vector norm in $\mathbb{R}^n$ at each time k and for discrete time signals $\|x_k\|_{2\delta} \triangleq$

$(\sum_{i=0}^k \delta^{k-i} x^T(i) x(i))^{1/2}$ defines the exponentially weighted l_2 norm of a vector function $x(k)$, where δ is a non-negative constant and k represents time. $x \in l_\infty$ means that $\|x\|_\infty = \sup_{k \geq 0} |x(k)|$ exists. $\lambda_{\max}(A)$ and $\lambda_{\min}(A)$ represent the maximum and minimum eigenvalue of a matrix A , respectively.

3. HVAC zone modeling and control objectives

We consider the dynamical model of heat transfer associated with the HVAC system in a typical building consisting of N thermal zones, as follows [3,49,50]:

$$\frac{dT_{z_i}}{dt} = \frac{f_{sa_i} \rho_a C_{pa}}{C_{z_i}} (T_{sa_i} - T_{z_i}) + \sum_{j \in M_i} \frac{U_{w_{i,j}} A_{w_{i,j}}}{C_{z_i}} (T_{w_{i,j}} - T_{z_i}) + \sum_{p \in N_i} \frac{f_{i,p} \rho_a C_{pa}}{C_{z_i}} (T_{z_p} - T_{z_i}) + \frac{K_{o_i}}{C_{z_i}} (T_o - T_{z_i}) + \frac{q_i}{C_{z_i}} \quad (1a)$$

$$\frac{dT_{w_{i,j}}}{dt} = \frac{U_{w_{i,j}} A_{w_{i,j}}}{C_{w_{i,j}}} (T_{z_i} - T_{w_{i,j}}) + \frac{U_{w_{i,j}} A_{w_{i,j}}}{C_{w_{i,j}}} (T_o - T_{w_{i,j}}) \quad (1b)$$

$$\frac{dT_{w_{i,j}}}{dt} = \frac{U_{w_{i,j}} A_{w_{i,j}}}{C_{w_{i,j}}} (T_{z_i} - T_{w_{i,j}}) + \frac{U_{w_{i,j}} A_{w_{i,j}}}{C_{w_{i,j}}} (T_{z_p} - T_{w_{i,j}}) \quad (1c)$$

$\forall i = 1, \dots, N$

Eq. (1a) states that the rate of change of the zone temperature T_{z_i} is affected by the supply air temperature, the temperature of surrounding surfaces, the temperature of neighboring zones, weather conditions and internal and external heat gains. The various terms in Eqs. (1a)–(1c) are listed in Table 1 and explained as follows: The temperature T_{z_i} in zone i is expressed in °C. C_{z_i} is the overall thermal capacity of the zone in J/°C. f_{sa_i} denotes the volume flow rate of the supply air in m³/s, while T_{sa_i} represents the supply air temperature in °C. C_{pa} represents the specific heat of air in J/kg°C and ρ_a denotes air density in kg/m³. The effect of surrounding surfaces temperature $T_{w_{i,j}}$ on zone temperature is shown with $U_{w_{i,j}}$ being the overall heat transfer coefficient of the j th surface that surrounds zone i in W/m²°C and $A_{w_{i,j}}$ denoting the respective surface area in m². M_i is the set of surfaces that surround zone i . T_{z_p} denotes the zone temperature for a neighboring

zone p and $f_{i,p}$ represents volume flow rate in m³/s between zones i and $p \in N_i$ due to convection. N_i is the set of neighboring zones of zone i . In addition, outside temperature is represented by T_o , with K_{o_i} denoting the respective thermal transfer coefficient in W/°C. Finally, q_i is used to model heat gains and losses, which may emerge due to human activity in the room, lights, electrical equipment, as well as plant and model mismatch, such as wall leakage. q_i is considered to be an unknown bounded disturbance that is either constant or slowly varying and is measured in Watt (W). It is assumed that the air in each zone i is fully mixed, zone distribution is uniform, air density is constant, there are no pressure losses across the zone and in the mixing section and Air Handling Units (AHUs) are ideal.

The temperature $T_{w_{i,j}}$ of the j th surface that surrounds zone i is affected by the temperature of the two spaces that it separates. One of the two spaces is the zone, while the other one is either another zone p or the outside. In the case of a surface separating a zone and the outside space, surface temperature satisfies the differential Eq. (1b), where T_o represents the outside temperature. For a surface that separates two zones, namely i and p , surface temperature satisfies the differential Eq. (1c). In (1b) and (1c), $C_{w_{i,j}}$ denotes the overall thermal capacity of the corresponding surface, $U_{w_{i,j}}$ is the overall heat transfer coefficient of the respective surface and is $A_{w_{i,j}}$ is the surface area in m². In (1c) T_{z_p} represents the temperature of the neighboring zone p that is separated by the surface.

The supply air from the air unit has a direct impact on zone temperature, while the temperature of the surfaces that surround a thermal zone is also a crucial factor due to heat transfer between air and the walls due to conduction. In addition, open surfaces between zones, such as open doors or windows, let heat transfer between zones due to convection and affect zone climate conditions significantly. Zone temperature is also affected by heat gains caused by human activity, lights, equipment, and weather conditions.

In the heat transfer model (1a)–(1c) it is assumed that all temperatures are measured. For a given building, identification techniques may be used to calculate the various constants that appear in (1a)–(1c). These parameters may change with time due to various effects described below which could make the initially estimated parameters invalid. For example, the opening of a door due to human activity leads to air flow, and thus, heat transfer between zones due to convection, as analyzed in [8], and may significantly change the parameters $f_{i,p}$ and A_{ij} in Eqs. (1a),(1c). In addition to that, leakages, material degradation and normal wear and tear may alter slowly but steadily thermal capacities, while such changes are not easily measured. The performance of insulation materials may degrade with time sometimes during the first 2–3 years after installation [51], which may lead to some parameter changes of up to 20%. Air density as well as zone temperature is considered to be uniformly distributed. However, depending on the air temperature, both the air density and the specific heat of air may slightly vary. A control system that is based on some initial estimate of the model parameters may considerably degrade in performance due to large parameter changes over time. In this paper, we design a controller that does not rely on the exact values of the parameters and has the capability to adapt to parameter changes. The structure of the model (1a) to (1c) motivates the use of a distributed controller structure where a controller is designed for each zone by taking into account possible interactions with neighboring zones in order to meet the objectives of each zone. This distributed structure has better robustness properties than a centralized one where a single central controller controls all zones. In the distributed case, a failure in a particular zone may be easily isolated, whereas in the centralized case it may affect all zones.

Table 1
Nomenclature.

Symbol	Definition
ρ_a	Air density in kg/m ³
$A_{w_{i,j}}$	Area of j th surface that surrounds zone i in m ²
C_{pa}	Specific heat of air in J/kg°C
$C_{w_{i,j}}$	Overall thermal capacity of j th surface that surrounds zone i in J/°C
C_{z_i}	Overall thermal capacity of i th zone in J/°C
f_{sa_i}	Volume flow rate of supply air of i th zone in m ³ /s
$f_{i,p}$	Volume flow rate between zones i and p due to convection in m ³ /s
K_{o_i}	Thermal transfer coefficient between zone i and the outside in W/°C
M_i	Set of surfaces that surround zone i
N	Total number of zones
N_i	Set of direct neighboring zones of i th zone
q_i	Internal heat gain of zone i in W
T_m	Target temperature for zone i in °C
T_o	Outside temperature in °C
T_s	Sampling time in s
T_{sa_i}	Supply air temperature in °C
$T_{w_{i,j}}$	Temperature of j th surface that surrounds zone i in °C
T_{z_i}	Temperature of i th zone in °C
$U_{w_{i,j}}$	Overall heat transfer coefficient of j th surface that surrounds zone i in W/m ² °C

Supply air is used as the control input to regulate zone temperature. HVAC systems can be divided into two major categories according to the way the adjust supply air. A large part of HVAC systems design focuses on appropriately adjusting supply air volume flow rate, while keeping supply air temperature constant to some predefined value. Such systems are known as Variable Air Volume (VAV) systems. On the other hand, there exist systems that keep supply air volume flow rate constant and adjust supply air temperature to meet the zone temperature objective. Such systems are known as Constant Air Volume (CAV) systems. Fan Coil Units (FCUs) are a widely applied type of HVAC equipment that often use constant air volume, or a small set of predefined values for supply air volume flow rate, and they operate by properly adjusting supply air temperature [47]. FCUs may often serve as the final air distribution units that adjust supply air temperature in multi-zone HVAC systems [52].

Control objective: Given the measurements T_{z_i} , $T_{w_{i,j}}$ $\forall j \in M_i$, T_{z_p} $\forall p \in N_i$ and T_o design the supply air temperature T_{sa_i} for each zone i , so that zone temperature T_{z_i} reaches a desired constant temperature target T_{m_i} , specified by the users of the system. In our formulation, we assume constant supply air volume flow rate, as in [49], and the control input is supply air temperature.

4. Distributed control scheme for multi-zone HVAC systems

4.1. Structure of control scheme

Let's start by defining the zone temperature tracking error $e_i = T_{z_i} - T_{m_i}$ $\forall i = 1, \dots, N$ which denotes how far the zone temperature deviates from the desired one. The control objective is to drive this error to zero or as close to zero as possible for all zones. Based on (1a), the tracking error satisfies the following differential equation:

$$\dot{e}_i = -\alpha_i e_i + \frac{f_{sa_i} \rho_a C_{pa}}{C_{z_i}} T_{sa_i} + \sum_{j \in M_i} \frac{U_{w_{i,j}} A_{w_{i,j}}}{C_{z_i}} T_{w_{i,j}} - \alpha_i T_{m_i} + \sum_{p \in N_i} \frac{f_{i,p} \rho_a C_{pa}}{C_{z_i}} T_{z_p} + \frac{K_{o_i}}{C_{z_i}} T_o + \frac{q_i}{C_{z_i}} \quad \forall i = 1, \dots, N \quad (2)$$

where the parameter:

$$\alpha_i = \frac{f_{sa_i} \rho_a C_{pa} + \sum_{p \in N_i} f_{i,p} \rho_a C_{pa} + \sum_{j \in M_i} U_{w_{i,j}} A_{w_{i,j}} + K_{o_i}}{C_{z_i}}, \quad \forall i = 1, \dots, N \quad (3)$$

indicates how heat flows in the zone as a result of all contributing factors, such as zone dynamics, neighboring zones and surrounding surfaces.

The discrete-time version of (2), which will be used to design the control input for digital implementation with sampling period T_s , is given as:

$$e_i(k+1) = \alpha_{d_i} e_i(k) + b_{d_i} \left[T_{sa_i}(k) + \sum_{j \in M_i} \frac{U_{w_{i,j}} A_{w_{i,j}}}{f_{sa_i} \rho_a C_{pa}} T_{w_{i,j}}(k) - \frac{\alpha_i C_{z_i}}{f_{sa_i} \rho_a C_{pa}} T_{m_i} + \sum_{p \in N_i} \frac{f_{i,p}}{f_{sa_i}} T_{z_p}(k) + \frac{K_{o_i}}{f_{sa_i} \rho_a C_{pa}} T_o(k) + \frac{1}{f_{sa_i} \rho_a C_{pa}} q_i(k) \right] \quad \forall i = 1, \dots, N \quad (4)$$

where $e(k)$ is the value of $e(t)$ at time $t = kT_s$ and $\forall i = 1, \dots, N$:

$$\alpha_{d_i} = e^{-\alpha_i T_s} \quad (5a)$$

$$b_{d_i} = \frac{1}{\alpha_i} (1 - e^{-\alpha_i T_s}) \frac{f_{sa_i} \rho_a C_{pa}}{C_{z_i}} \quad (5b)$$

The control input T_{sa_i} $\forall i = 1, \dots, N$ should be chosen, so that error e_i $\forall i = 1, \dots, N$ converges to zero or close to zero despite the effects of neighboring zones, surrounding surfaces, outside weather conditions, etc. This reasoning suggests the following structure for the controller of each zone $\forall i = 1, \dots, N$:

$$T_{sa_i}(k) = -K_{e_i}^* e_i(k) - \sum_{j \in M_i} K_{w_{i,j}}^* T_{w_{i,j}}(k) - K_{m_i}^* T_{m_i} - \sum_{p \in N_i} K_{i,p}^* T_{z_p}(k) - K_{out_i}^* T_o(k) - h_i G_i(k) \quad (6a)$$

$$G_i(k) = G_i(k-1) + e(k) \quad (6b)$$

where h_i is the gain of the accumulator in (6b) and the gains $K_{e_i}^*$, $K_{w_{i,j}}^*$, $K_{m_i}^*$, $K_{i,p}^*$ and $K_{out_i}^*$ are calculated as follows:

$$K_{e_i}^* = \frac{\alpha_{m_i} - \alpha_{d_i}}{b_{d_i}}, \quad K_{w_{i,j}}^* = \frac{U_{w_{i,j}} A_{w_{i,j}}}{f_{sa_i} \rho_a C_{pa}}, \quad K_{m_i}^* = -\frac{\alpha_i C_{z_i}}{f_{sa_i} \rho_a C_{pa}}, \quad (7)$$

$$K_{i,p}^* = \frac{f_{i,p}}{f_{sa_i}}, \quad K_{out_i}^* = \frac{K_{o_i}}{f_{sa_i} \rho_a C_{pa}}$$

where α_{m_i} is a design parameter which, together with gain h_i , will be selected to achieve certain stability properties for the zone temperature.

By applying the proposed control input (6a) on zone i , (4), the temperature tracking error of each zone satisfies the following error equation:

$$e_i(k+1) = \alpha_{m_i} e_i(k) + b_{d_i} \left(-h_i G_i(k) + \frac{1}{f_{sa_i} \rho_a C_{pa}} q_i(k) \right) \quad (8)$$

$\forall i = 1, \dots, N$. From (8), we derive that the temperature tracking error is given by the following transfer function:

$$e_i(z) = \frac{z-1}{z^2 + (h_i b_{d_i} - 1 - \alpha_{m_i})z + \alpha_{m_i}} \frac{b_{d_i}}{f_{sa_i} \rho_a C_{pa}} q_i(z) \quad (9)$$

where z is the z -transform. The desired characteristic equation for (9) that guarantees stability and a certain desired rise time, overshoot and settling time [53] is of the form $z^2 + c_1 z + c_0 = 0$, which can be used to choose the parameters h_i and α_{m_i} . Given the stability of the characteristic equation of (9), the zone temperature tracking error converges to zero exponentially fast for constant heat gains q_i .

The controller described by (6a), (6b) and (7) assumes that the parameters of the model are known exactly which, as we argued before, is not a realistic assumption. Therefore, the significance of the controller (6a)–(7) which drives the errors to zero is to serve as a reference when we consider the case of unknown model parameters in the following section.

4.2. Estimation of controller gains

Since the parameters $K_{e_i}^*$, $K_{w_{i,j}}^*$, $K_{m_i}^*$, $K_{i,p}^*$ and $K_{out_i}^*$ are unknown, the controller (6a)–(7) cannot be implemented. In this case, we use the following control input:

$$T_{sa_i}(k) = -K_{e_i}(k) e_i(k) - \sum_{j \in M_i} K_{w_{i,j}}(k) T_{w_{i,j}}(k) - K_{m_i}(k) T_{m_i} - \sum_{p \in N_i} K_{i,p}(k) T_{z_p}(k) - K_{out_i}(k) T_o(k) - K_i(k) \quad (10)$$

where the gains $K_{e_i}(k)$, $K_{w_{i,j}}(k)$ $\forall j \in M_i$, $K_{m_i}(k)$, $K_{i,p}(k)$ $\forall p \in N_i$, $K_{out_i}(k)$ are replacing the gains $K_{e_i}^*$, $K_{w_{i,j}}^*$, $K_{m_i}^*$, $K_{i,p}^*$, $K_{out_i}^*$, respectively. The accumulator $h_i G_i(k)$ has been replaced with $K_i(k)$, with K_i^* being the unknown desired value of $K_i(k)$.

The updating rules of the gains $K_{e_i}(k)$, $K_{w_{i,j}}(k)$, $K_{m_i}(k)$, $K_{i,p}(k)$, $K_{out_i}(k)$ and $K_i(k)$ in (10) are developed according to the following process. First, the zone temperature tracking error Eq. (4) is

re-written as:

$$e_i(k+1) = \alpha_{m_i} e_i(k) + b_{d_i}^* \left(T_{sai}(k) + K_{e_i}^* e_i(k) + \sum_{j \in M_i} K_{w_{i,j}}^* T_{w_{i,j}}(k) + K_{m_i}^* T_{m_i} + \sum_{p \in N_i} K_{i,p}^* T_{z_p}(k) + K_{out_i}^* T_o(k) + K_i^* \right) \quad (11)$$

which is expressed in the form of the following linear parametric model [54]:

$$z_i(k) = \Theta_i^* \Phi_i(k) \quad (12)$$

where

$$z_i(k) = e_i(k) - \alpha_{m_i} e_i(k-1) \quad (13a)$$

$$\begin{aligned} \Theta_i^* &= [b_{d_i}^*, b_{d_i}^* K_{e_i}^*, b_{d_i}^* [K_{w_{i,j}}^*]_{j \in M_i}, b_{d_i}^* K_{m_i}^*, b_{d_i}^* [K_{i,p}^*]_{p \in N_i}, \\ &\quad \times b_{d_i}^* K_{out_i}^*, b_{d_i}^* K_i^*]^\top \\ &= [\theta_0^*, \theta_1^*, [\theta_{2,i,j}^*]_{j \in M_i}, \theta_3^*, [\theta_{4,i,p}^*]_{p \in N_i}, \theta_5^*, \theta_6^*]^\top \end{aligned} \quad (13b)$$

$$\begin{aligned} \Phi_i(k) &= [T_{sai}(k-1), e_i(k-1), [T_{w_{i,j}}(k-1)]_{j \in M_i}, T_{m_i}, \\ &\quad \times [T_{z_p}(k-1)]_{p \in N_i}, T_o(k-1), 1]^\top \end{aligned} \quad (13c)$$

The linear in the parameters model (12) is used to estimate Θ_i^* . Let

$$\begin{aligned} \Theta_i(k) &= [\theta_0(k), \theta_1(k), [\theta_{2,i,j}(k)]_{j \in M_i}, \theta_3(k), [\theta_{4,i,p}(k)]_{p \in N_i}, \\ &\quad \times \theta_5(k), \theta_6(k)]^\top \end{aligned} \quad (14)$$

be the estimate of Θ_i^* at instant k . Then, the estimates of the controller gains are generated as follows $\forall i = 1, \dots, N$, $\forall j \in M_i$ and $\forall p \in N_i$:

$$\begin{aligned} K_{e_i}(k) &= \frac{\theta_1(k)}{\theta_0(k)}, \quad K_{w_{i,j}}(k) = \frac{\theta_{2,i,j}(k)}{\theta_0(k)}, \quad K_{m_i}(k) = \frac{\theta_3(k)}{\theta_0(k)}, \\ K_{i,p}(k) &= \frac{\theta_{4,i,p}(k)}{\theta_0(k)}, \quad K_{out_i}(k) = \frac{\theta_5(k)}{\theta_0(k)}, \quad K_i(k) = \frac{\theta_6(k)}{\theta_0(k)} \end{aligned} \quad (15)$$

where

$$\tilde{\Theta}_i(k) = \Theta_i(k-1) + T_s \Gamma_i e_i(k) \Phi_i(k) \quad (16a)$$

$$\begin{aligned} [\Theta_i(k)]_s &= \begin{cases} [\tilde{\Theta}_i(k)]_s & \text{if } [L_i]_s \leq [\tilde{\Theta}_i(k)]_s \leq [U_i]_s \\ [L_i]_s & \text{if } [\tilde{\Theta}_i(k)]_s < [L_i]_s \\ [U_i]_s & \text{if } [\tilde{\Theta}_i(k)]_s > [U_i]_s \end{cases} \\ \forall s &= 0, \dots, 4 + |M_i| + |N_i| \end{aligned} \quad (16b)$$

$$e_i(k) = \frac{z_i(k)}{m_i^2(k)} = \frac{e_i(k) - \alpha_{m_i} e_i(k-1)}{m_i^2(k)} \quad (16c)$$

$$m_i^2(k) = 1 + T_s \Phi_i^\top(k) \Gamma_i \Phi_i(k) \quad (16d)$$

where $[\Theta_i(k)]_s$ denotes the s th element of $\Theta_i(k)$ and the diagonal matrix $\Gamma_i = \text{diag}(\gamma_{b_{d_i}}, \gamma_{e_i}, [\gamma_{w_{i,j}}]_{j \in M_i}, \gamma_{m_i}, [\gamma_{z_p}]_{p \in N_i}, \gamma_{o_i}, \gamma_i)$ consists of the positive design constants $\gamma_{b_{d_i}}, \gamma_{e_i}, [\gamma_{w_{i,j}}]_{j \in M_i}, \gamma_{m_i}, [\gamma_{z_p}]_{p \in N_i}, \gamma_{o_i}, \gamma_i$, referred to as adaptive gains, that affect the speed of adaptation. U_i, L_i are upper and lower bounds for the values of the elements of $\Theta_i(k)$ and $[L_i]_0 > 0$. The lower bound $[L_i]_0$ of $\theta_0(k)$ has to be strictly positive to guarantee boundedness of the estimated gains. However, this is not constricting, since the real value of $\theta_0(k)$ is strictly positive

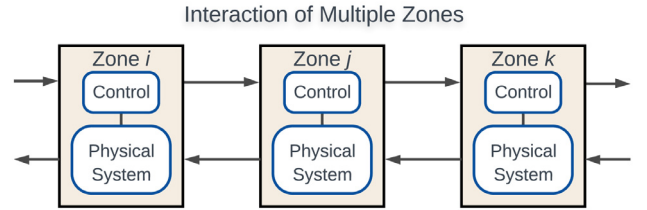
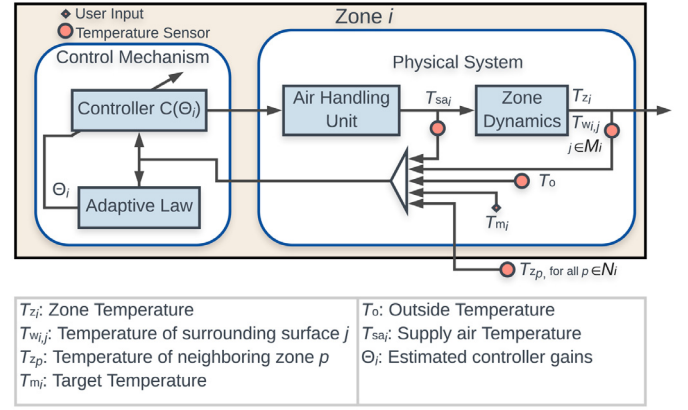


Fig. 1. Zone Control Diagram with Adaptation.

due to system structure. The rest of the bounds $[U_i]_0$ and $[U_i]_s$, $[L_i]_s \forall s = 1, \dots, 4 + |M_i| + |N_i|$ serve as conservative bounds for the gain estimates to guarantee the boundedness of the gains in Eq. (15). Signals e_i and m_i are auxiliary signals to be calculated inside the control mechanism.

The controller (10) with the adaptation law (15)–(16d) form the adaptive control system which does not require the knowledge of the model parameters. The following Theorem describes the stability properties of the closed loop system.

Theorem 1. Consider the overall closed-loop system composed of the N subsystems (4) using the control scheme (10), (15). Then, $\forall i \in I_\infty$ and e_i converges to zero as $k \rightarrow \infty$.

Proof. The proof of Theorem 1 is presented in the Appendix. $\square$

Theorem 1 guarantees that all signals in the closed loop system of each zone are bounded and zone temperature tracking errors converge to zero with time. Thus, the scheme is able to completely cancel the impact of neighboring zones, surrounding surfaces, outside temperature and internal and external heat gains q_i on zone temperature, making zones reach their temperature objectives satisfactorily. The result holds for unknown constant system parameters and disturbances, but can also be extended to slowly varying parameters or parameters with infrequent finite changes [55]. The proposed control scheme is scalable, since it distributes computational burden and its stability and convergence properties do not depend on the building and HVAC size. It should be noted that the closed loop poles $\alpha_{m_i} \in (0, 1)$ should be chosen according to the individual thermal zone's requirements to achieve fast temperature regulation, without exhibiting unwanted oscillations or overshoot. Fig. 1 shows the closed loop configuration of one zone and interaction with other zones.

In the next section, we use simulations to demonstrate the results of Theorem 1 and evaluate the properties and performance of the proposed control scheme.

5. Simulation results and discussion

In this section, we present two case studies to show the effectiveness of the proposed control scheme. In the first case study, we

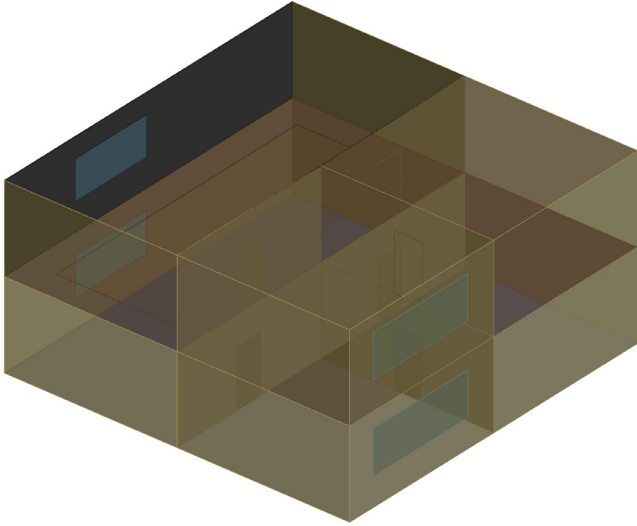


Fig. 2. Case study I: Model of six-zone building.

apply the proposed scheme to regulate temperature in a two-story building with six interconnected thermal zones and we demonstrate the effect on temperature tracking, focusing on the impact of abrupt parameter changes. In the next study, we apply the proposed scheme to a large primary school building model and we examine the performance of the proposed control scheme under different scenarios. Since one of the basic characteristics of the proposed scheme is the utilization of learning, in both cases we compared its performance to the performance of a distributed control scheme that does not include adaptive learning. Both systems are simulated using EnergyPlus for the building and HVAC model and Matlab/Simulink for the control scheme, co-simulated using the Buildings Control Virtual Test Bed (BCVTB).

5.1. Case study I: Building with six interconnected thermal zones

5.1.1. Building description

For our first study, we consider a building with two stories, each of which is divided into three thermal zones, as shown in Fig. 2. The building includes zones that may be separated by walls but also zones that have no wall separation, to show how the proposed algorithm can handle both types of interconnections between zones. Each story has a large zone on the north side with dimensions $12.092 \times 6.096 \times 3.048 \text{ m}^3$ and two smaller zones, one on the west side and one on the east side that have the same size and dimensions $6.096 \times 6.096 \times 3.048 \text{ m}^3$. Any given zone is connected with a neighboring zone via a door of dimension $1 \times 2 \text{ m}^2$. The north zone has a window of dimensions $3 \times 1.333 \text{ m}^2$, while the west zone has a window of dimension $4 \times 1.5 \text{ m}^2$. In addition, there is a permanent horizontal opening of dimensions $10.192 \times 4.096 \text{ m}^2$ which connects the north zones of the two stories. The HVAC equipment is considered to be ideal, thus able to provide supply air with the desired temperature and volume flow rate. Supply air temperature is going to be calculated by the control scheme, while supply air volume flow rates are predefined. The north zones have supply air volume flow rate $0.260 \text{ m}^3/\text{s}$ and the west and east ones have supply air volume flow rate $0.220 \text{ m}^3/\text{s}$. Heat gains in each zone consist of heat gains from occupants, lights and electronic equipment and they are scheduled to vary during the day. The control system is implemented using sampling time $T_s = 60\text{s}$.

5.1.2. Simulation results

For the purpose of this demonstration, we consider a scenario where the HVAC system is operating, when at 8:00 am doors open and heat gain increases as people start using the building. Open doors result in direct interactions between zones. Several other weather, internal gains and parameter changes occur during the day.

The proposed distributed adaptive control scheme regulates the temperature in all zones and its performance is compared to a distributed constant-gain control scheme which assumes exact knowledge of the parameters initially without being aware of any parameter changes. In order to quantify the impact of the proposed scheme, we propose the following metrics, which correspond to the change in the overall tracking error of the system and the overall change in energy consumption when compared to the baseline controller:

$$\epsilon_z = \sum_{i=1}^N \left[\frac{\sum_k |e_{z_i}(k)| - \sum_k |\tilde{e}_{z_i}(k)|}{\sum_k |e_{z_i}(k)|} \right] \quad (17)$$

$$\epsilon_E = \frac{|E_{\text{HVAC}}^N| - |\tilde{E}_{\text{HVAC}}^N|}{|E_{\text{HVAC}}^N|} \quad (18)$$

for total operating time where e_{z_i} and E_{HVAC}^N denote the zone temperature tracking error of zone i and the HVAC energy consumption in all N zones, respectively, for the baseline controller, while $\tilde{e}_{z_i}$ and $\tilde{E}_{\text{HVAC}}^N$ denote the performance and energy consumption, respectively, of the proposed distributed adaptive control scheme. For the current selection of design constants, the proposed control scheme achieves 28.35% improvement in overall temperature tracking performance over a one-month period. Based on the EnergyPlus energy consumption reporting, such improvement may be achieved with 2.05% reduction in total energy consumption.

Fig. 3 shows the behavior of each zone at 8:00am when each control scheme is applied. The proposed adaptive scheme is able to react faster to the changes in the system and compensate faster for the heat transfer in most of the zones, even though both schemes eventually reach the temperature goal. As shown in Fig. 3 and reflected in the reduction of the overall tracking error, in most thermal zones the temperature may reach the desired one, within a 2% error margin, several minutes faster when the adaptive scheme is applied, compared to the constant-gain scheme.

The fast reaction of the proposed control scheme is due to the adaptation of the controller gains to the new dynamics. For instance, Fig. 4 shows how the gain K_{m_i} changes to accommodate for the different dynamics due to a door opening and introduction of heat gains due to human activity. As shown in Eq. (7), the ideal value of K_{m_i} depends on the parameter α_i , which is defined in Eq. (3) and depends on inter-zone air flow due to convection and separating surface area. Thus, when a change occurs, the gain estimate of K_{m_i} adapts to capture the new dynamics and the gain adaptation results in more accurate supply air temperature to be applied to the zone. In the same figure, it is also shown how capturing parameter changes by the adaptive control scheme is reflected on the applied control input T_{sa_i} . The proposed learning-based scheme adjusts supply air temperature faster to the appropriate value. In several cases fast and accurate temperature regulation is crucial, such as in intensive care units in hospitals or in museums with sensitive exhibits [42]. Satisfactory temperature regulation may also have an impact on learning ability in school [56]. Fig. 5 shows the performance of the proposed scheme compared to a non-learning one for one full day of operation, and how the scheme adapts to several heat gain and parameter changes.

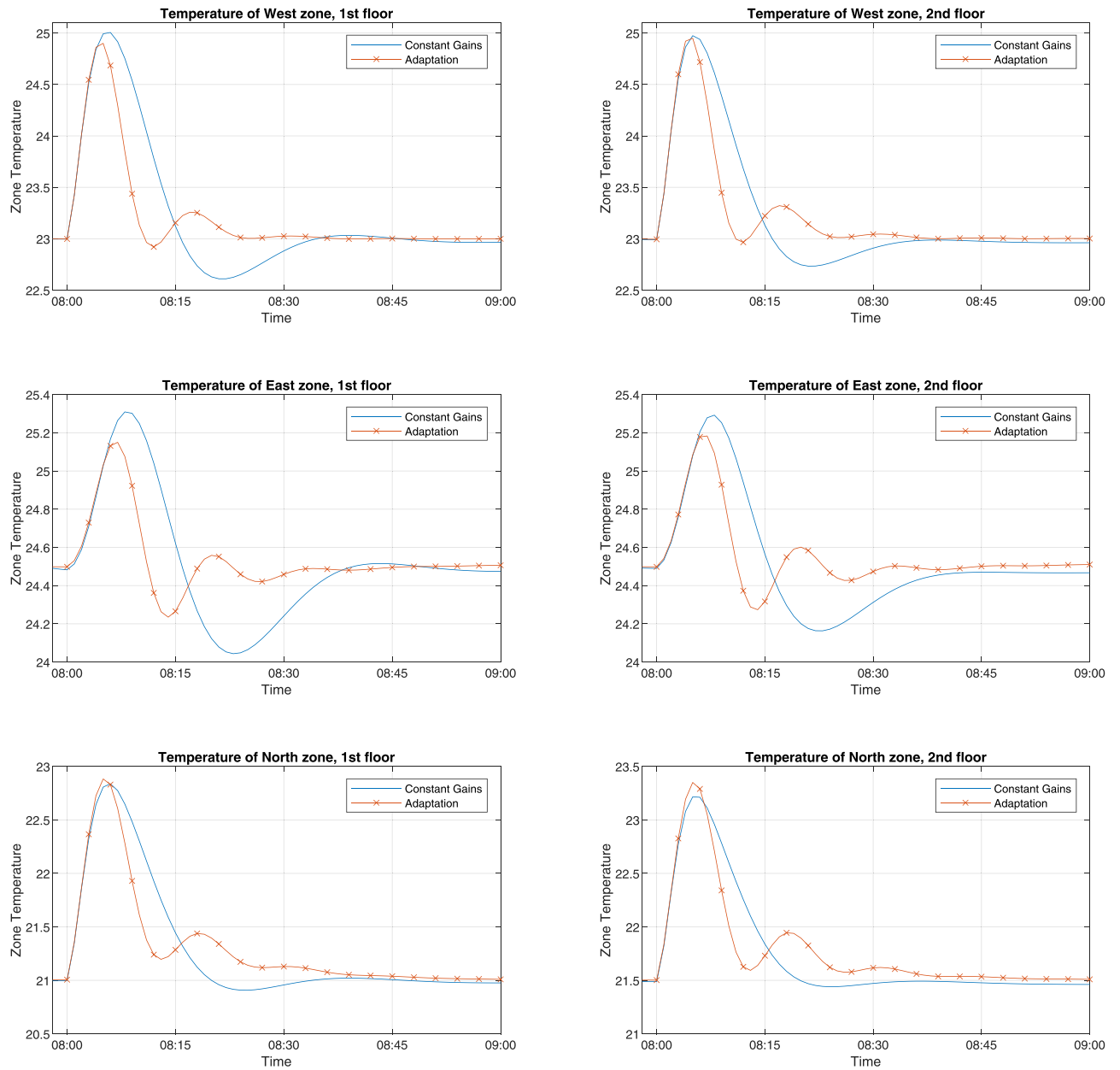


Fig. 3. Case study I: Temperature tracking in all zones.

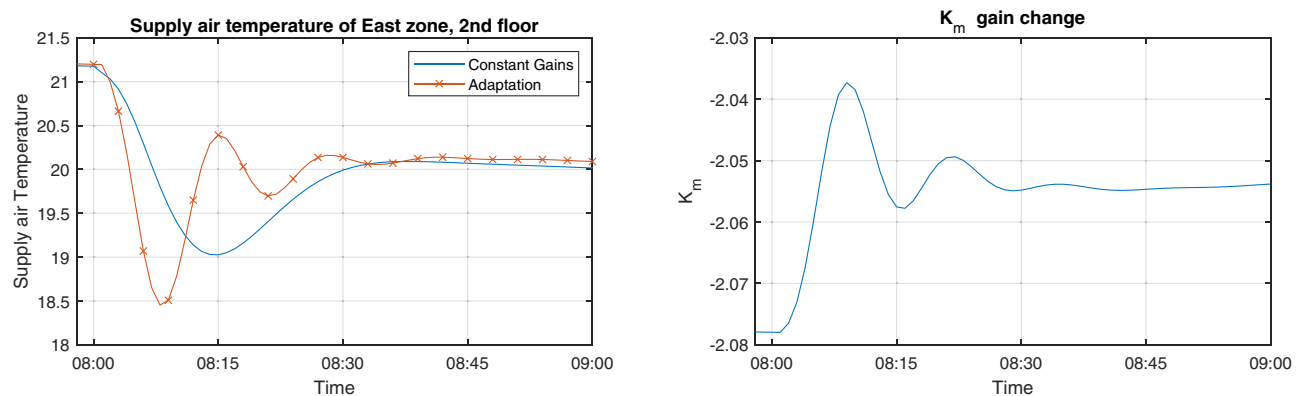


Fig. 4. Case study I: Supply air temperature and gain adaptation for East Zone, 2nd floor.

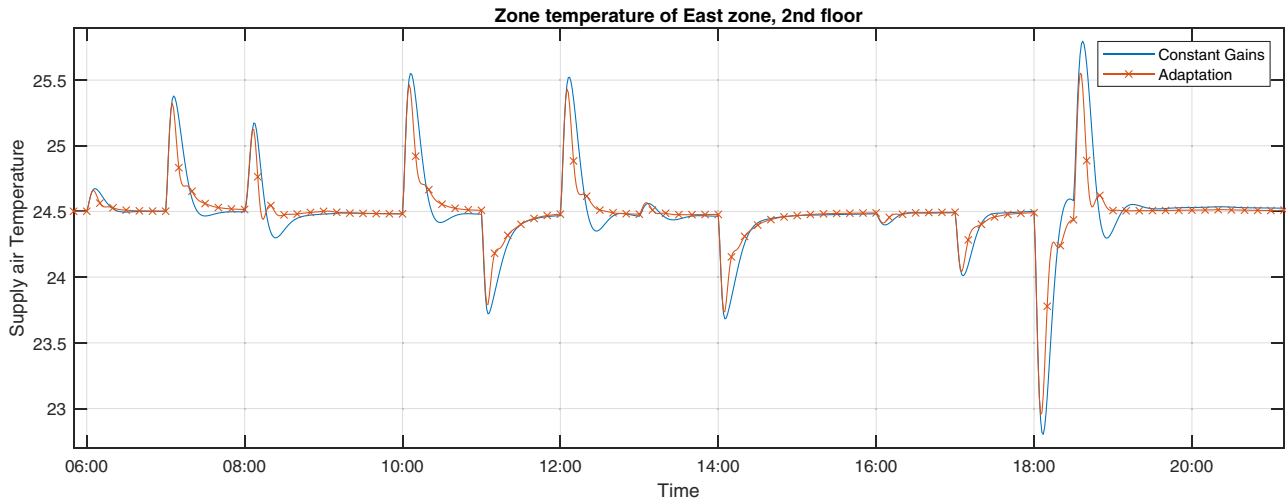


Fig. 5. Case study I: East Zone Temperature, 2nd floor.

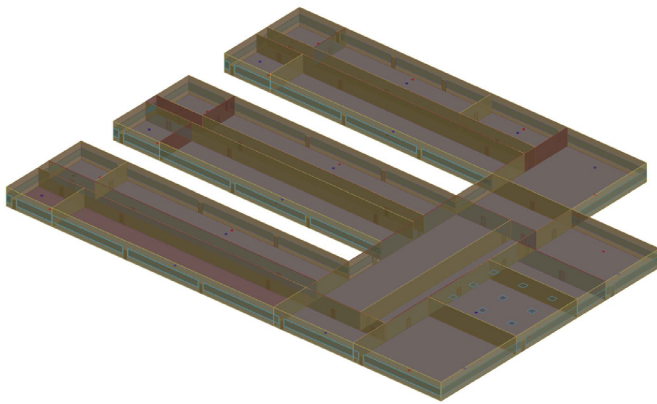


Fig. 6. Case study II: Model of primary school building.

5.2. Case study II: primary school building

5.2.1. Building description

The second case study refers to the application of the proposed scheme to a large building with multiple thermal zones and varying heat loads, in order to further demonstrate its performance. The algorithm is implemented and regulates temperature in a prototype primary school building model, which is the ANSI/ASHRAE/IES Standard 90.1-2016 Primary School model, located in Denver, from the ASHRAE Standard 90.1 prototype buildings suite, as developed by Pacific Northwest National Laboratory [57]. The school building has 25 thermal zones, which vary in size and use. According to their use, the zones have different heat load patterns, which is a result of different occupancy levels and human activity, lighting or equipment, as well as weather impact. Naturally, neighboring zones are physically interconnected via walls and doors. The 3D building model of the school building is presented in Fig. 6. The HVAC equipment of the building is ideal and is designed so supply air temperature may be controlled, while a constant value for the supply air volume flow rate is passed. Supply air volume flow rates are chosen based on zone dimensions. The modified EnergyPlus input data file (.idf) that describes the building and HVAC system is presented in the following Github link [58]. All schedules for occupancy levels, electrical equipment operation, lighting, door opening as well as weather conditions are included in the .idf file.

5.2.2. Simulation results

The HVAC system of the school building is operating only on weekdays from 7am to 5pm. All zones have their individual occupancy schedule that corresponds to their use and corresponding schedules for their equipment and lighting. Doors are scheduled to open and close at several times to capture changes in the ways neighboring zones are connected. We have selected the temperature target for all zones to be $T_{m_i} = 23^\circ\text{C}$. The control system is implemented using sampling time $T_s = 60\text{s}$. The design constants correspond to the desired closed loop poles for each zone and the learning rates of the adaptive laws. The closed loop poles are chosen so that stability is preserved and the system achieves temperature regulation fast and unwanted oscillations or overshoot are avoided. Learning rates reflect the system disturbances and possible parameter changes. The projection bounds of the adaptive laws are conservative bounds for the controller gains that are estimated on-line. For comparison, we have implemented the distributed control scheme with adaptive learning and a distributed control scheme that does not include adaptation.

The performance of the proposed algorithm is examined under different scenarios. First, we show the temperature tracking for the school's thermal zones during one day of operation during January. During one normal day of operation, the system has to deal with varying internal heat loads as well as interconnection changes due to doors. The tracking performance of ten different thermal zones during one day of operation is presented in Fig. 7. By simulating and evaluating the performance for the full month of January, we measured 36.80% improvement in temperature tracking and 13.68% improvement in overall energy consumption, as reported by EnergyPlus software, using metrics in Eq. (17) and Eq. (18). The corresponding input signals for the same day of operation are presented in Fig. 8, and they are all well-behaved. As shown in the figures, the system reacts whenever changes occur, with the proposed scheme being able to adapt to the changes faster, due to the existence of learning.

Through on-line learning and improved heat allocation between zones, the proposed control methodology has shown to improve both temperature tracking and energy consumption. According to design priorities, the system moderator may choose to give priority to improved temperature tracking or improved energy consumption, by changing the learning rate of the controller. We have chosen learning rates Γ_i by considering the nominal values of the building parameters to adapt the estimated parameters according to their scale. In order to illustrate how the choice of learning

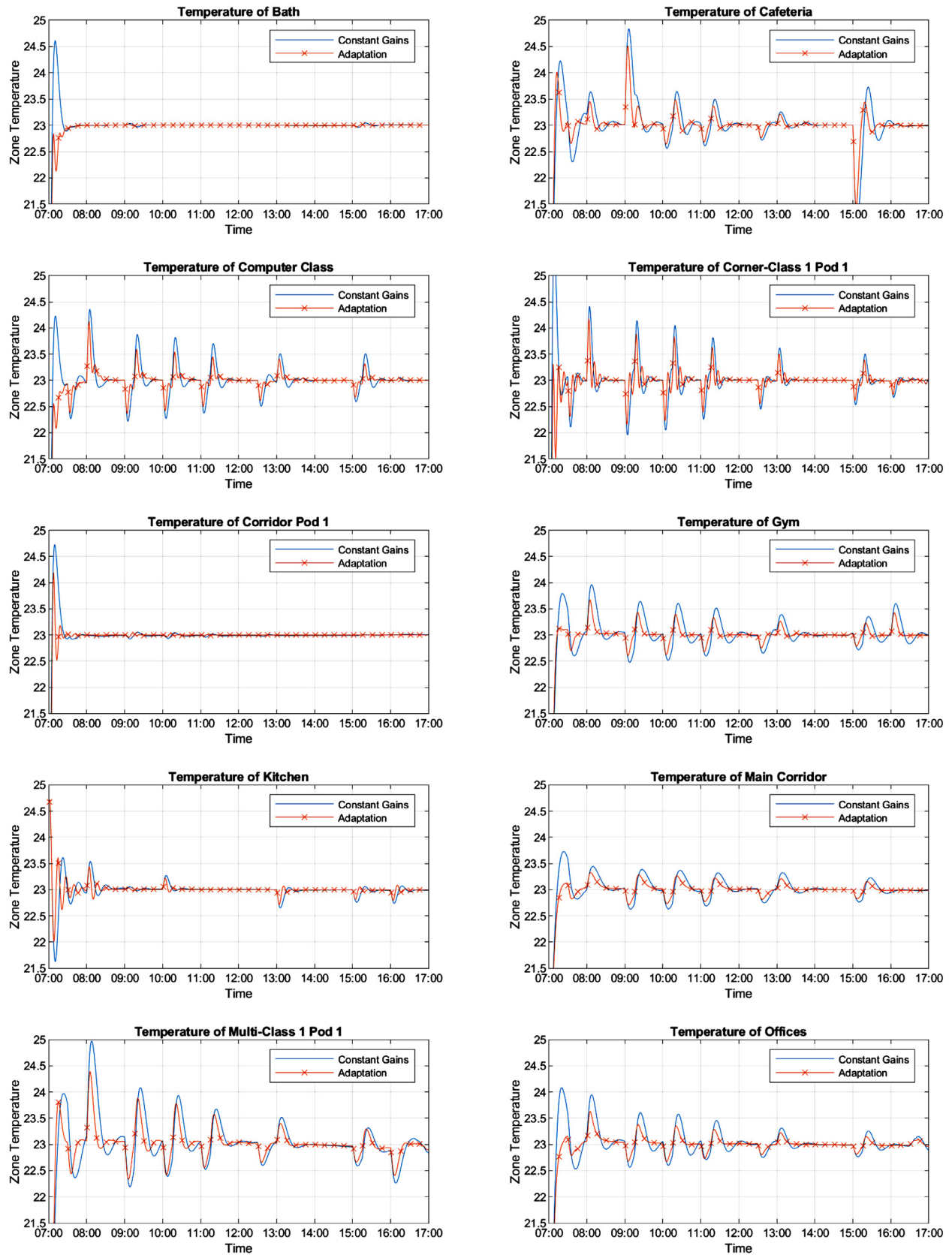


Fig. 7. Case study II: Temperature tracking in 10 different zones.

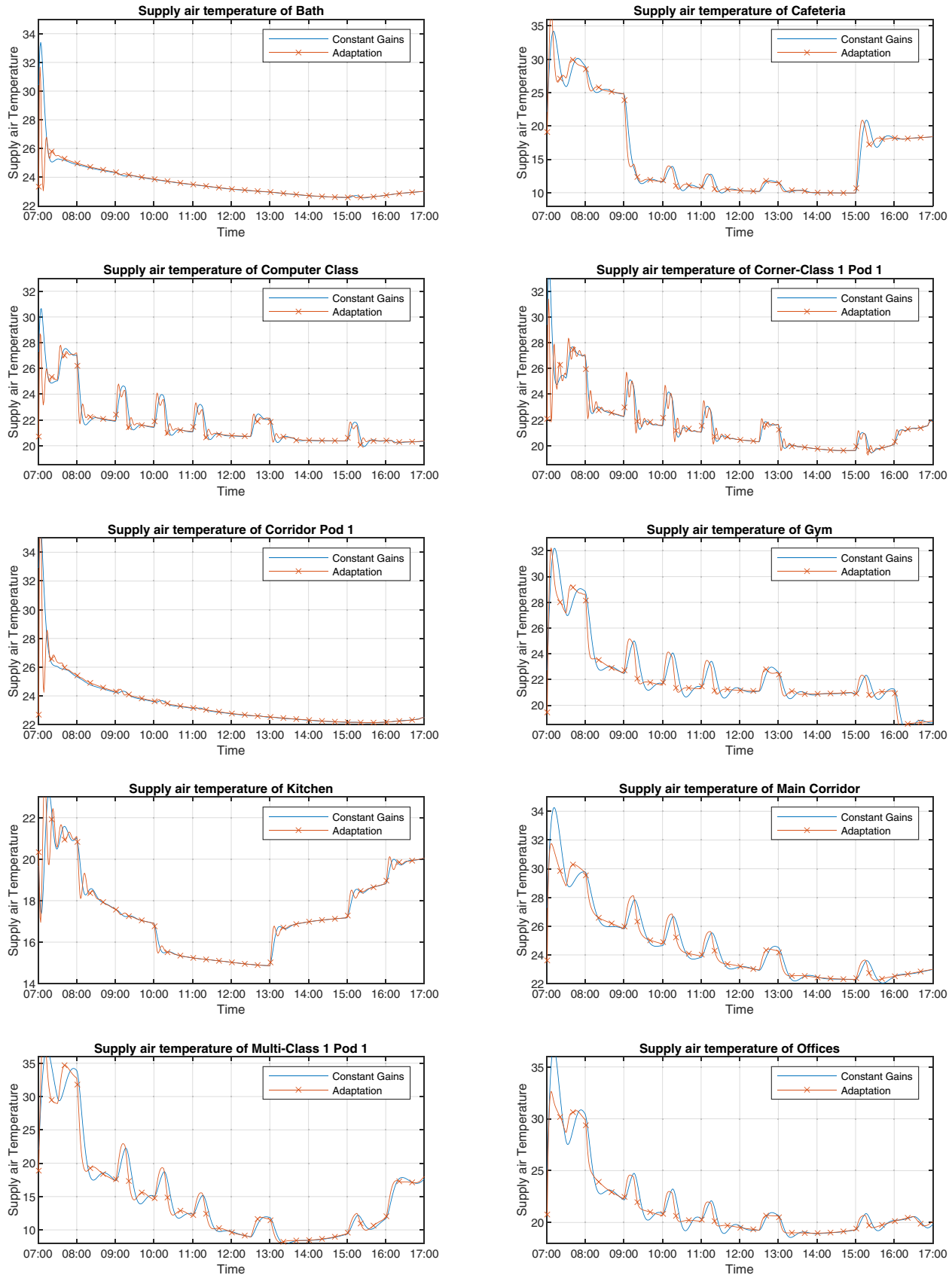


Fig. 8. Case study II: Supply air temperature of 10 zones.

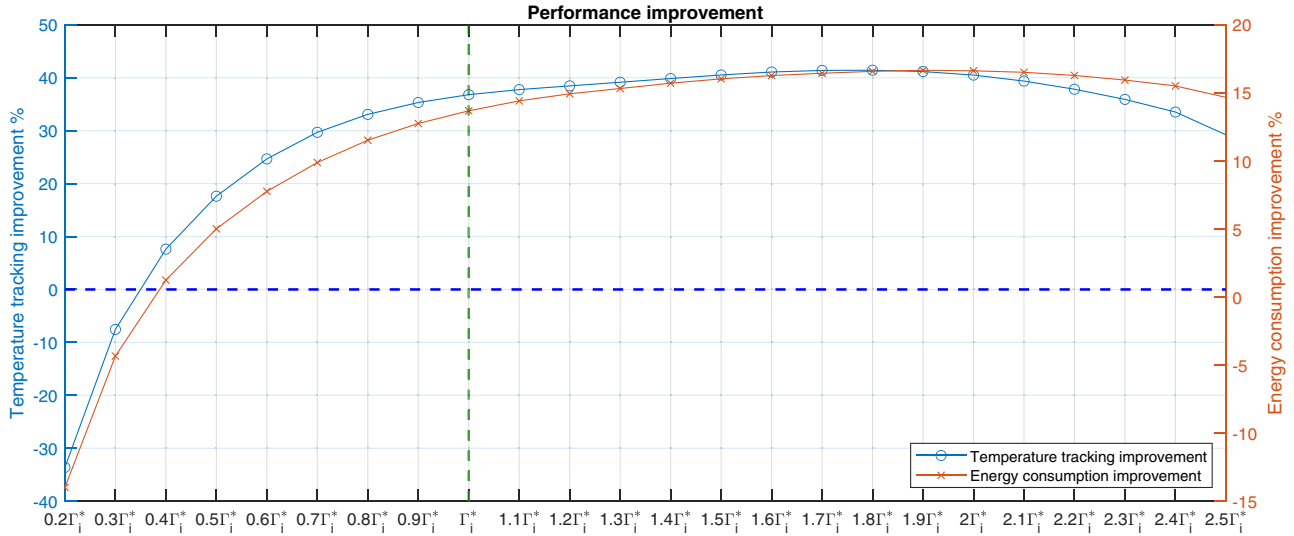


Fig. 9. Case study II: Improved temperature tracking and energy consumption based on the choice of Γ_i .

rate affects the trade-off between improved energy consumption and temperature tracking, we have simulated the system during the month of January, adjusting the value of Γ_i . Fig. 9 shows how temperature tracking and energy consumption change depending on the learning rate, considering the original choice of Γ_i from the previous simulation as the baseline and denoting it as Γ_i^* . Aligning with adaptive control theory, learning rate Γ_i should be chosen adequately large to be able to respond fast enough to changes and the varying load demand, but also not too large, in order to avoid undesired oscillations, which may deteriorate system performance and negatively affect HVAC equipment. This implies that learning rates should be selected at the neighborhood where temperature tracking and energy consumption become less sensitive to changes in the value of Γ_i . It should be noted that the performance recorded in Fig. 9 corresponds to the same one-month period and, thus, same weather conditions and heat load patterns, and may change for different time periods. Therefore, for the choice of Γ_i , the controller should be able to learn adequately fast the varying load demand, to provide improved performance throughout the year.

Every month of the year corresponds to different weather conditions and, thus, different loads to be satisfied by the HVAC system. Therefore, we test the system's performance during different periods of the year to capture the response of the control scheme during different weather conditions. The overall system is simulated during an one-year period. During one year, the average improvement is 34.57% in temperature tracking and 9.51% in overall energy consumption. Table 2 shows the results for each month and it can be shown that performance is improved, yet different for different weather conditions. The proposed control scheme is learning the needs of the building for different system changes and weather conditions, and is also able to continuously adapt to new conditions, since its learning ability is dynamic. Fig. 10 presents the controller gains for the Corner Class 1, Pod 1, during a one-year simulation, where it is shown how controller gains adjust for the different needs of each time period.

Some parameter changes may happen gradually in a building's life. Thermal and insulation properties of materials and equipment may degrade significantly after a few years [51]. Therefore, we simulate system performance during January, but after a few years, considering insulation degradation of 15%, according to [51]. Since

Table 2

Performance improvement when learning is introduced.

	Tracking error improvement	Energy consumption improvement
January	36.80%	13.68%
February	36.52%	14.03%
March	35.65%	11.79%
April	35.35%	12.39%
May	35.54%	6.54%
June	37.35%	5.19%
July	37.90%	7.28%
August	36.55%	7.04%
September	34.22%	4.52%
October	31.77%	9.01%
November	30.63%	10.18%
December	30.62%	12.81%
Full Year	34.57%	9.51%

the impact of material degradation is hard to measure and quantify, the gains of the baseline controller without adaptation are calculated using the nominal values of the material properties. If insulation properties degrade, we measured 22.76% improvement in temperature tracking and 7.79% improvement in overall energy consumption when using the proposed distributed adaptive control scheme compared to a scheme that calculates control gains based on the nominal values of the materials. This result implies that not only performance is improved using the proposed scheme, but also the need for maintenance and recalibration of the control mechanism is reduced.

Finally, we examined the importance of considering interconnections in the control design. In this case, we simulated the system performance during one full month of operation keeping the learning properties of the algorithm, but removing the controller terms that correspond to the direct and indirect heat transfer from neighboring zones. We measured that taking into account the physical interconnections of neighboring zones into the control design results in 30.14% improvement in temperature tracking and 10.61% improvement in overall energy consumption. While information on the indirect and direct impact of neighboring zones is not necessary for stability, heat is allocated through all thermal zones more efficiently when local controllers utilize such information.

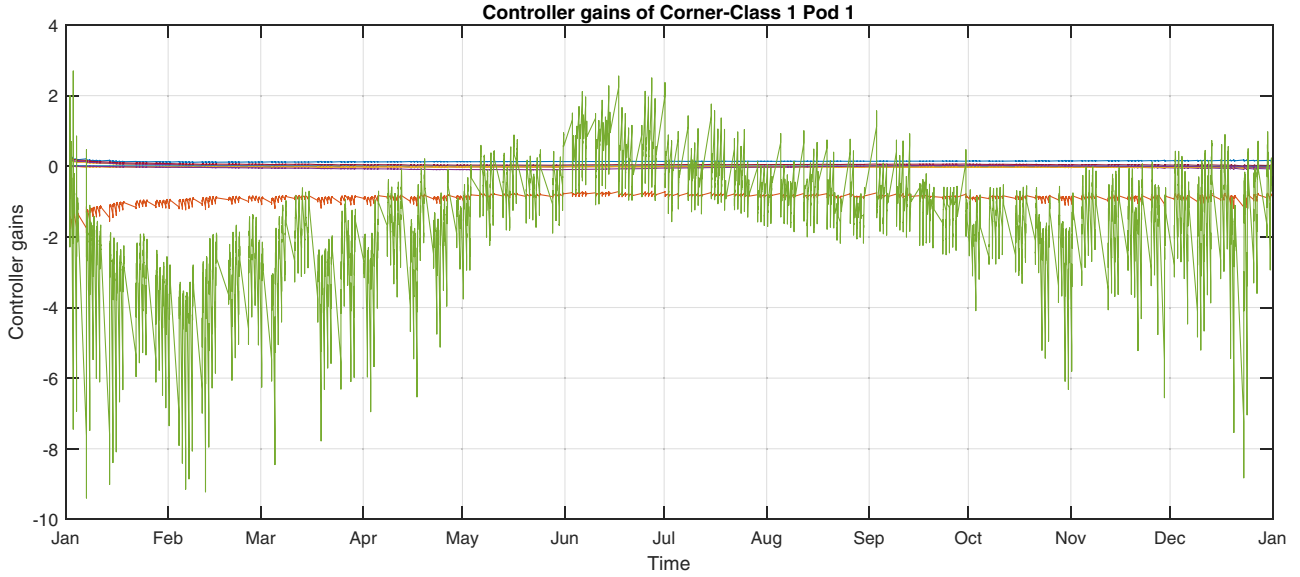


Fig. 10. Case study II: Adaptation of controller gains of CornerClass1Pod1 during 1 year.

6. Conclusions

In this paper we propose a distributed adaptive control scheme for temperature regulation in multi-zone HVAC systems. The proposed scheme allocates energy between thermal zones efficiently and does not require accurate knowledge of model parameters, thus adjusting to parameter changes and unknown disturbances. We showed that the proposed scheme guarantees bounded signals and convergence of zone temperature to the desired one for all building zones, while convergence properties do not depend on the size of the system, thus making the control scheme scalable. Performance is evaluated by simulating the behavior of two different buildings in EnergyPlus. The introduction of adaptive learning and exchange of information between zones is shown to provide faster response to changes and more accurate temperature control while consuming less energy in realistic scenarios, when human activity, lighting, electrical equipment, door opening and closing, and varying weather conditions are considered. The scheme has also been shown to retain better performance when material degradation is considered. Understanding the dynamic nature of buildings and uncertainty in HVAC system, as also noted in [59], and considering natural heat exchange between neighboring thermal zones in the control design has been shown to be an important step in more efficient and cost-effective temperature control.

Declaration of Competing Interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

Appendix. Proof of Theorem 1

Proof. $\Theta_i(k)$ is generated by Eqs. (16a),(16b), which can be written in the following compact form:

$$\Theta_i(k) = \text{Proj}[\Theta_i(k-1) + T_s \Gamma_i \epsilon_i(k) \Phi_i(k)] \quad (\text{A.19})$$

where we use the projection operator for discrete systems, as described in [54], with box constraints for each element of Θ_i .

The projection operator restricts adaptation between predefined bounds, within which the unknown desired gains are assumed to lie.

First, we establish the properties of m_i . From the definition of m_i , Eq. (16d), we derive the following inequality:

$$m_i^2(k) \geq 1 + T_s \lambda_{\min}(\Gamma_i) |\Phi_i(k)|^2 \quad (\text{A.20})$$

which implies:

$$\frac{|\Phi_i(k)|^2}{m_i^2(k)} \leq \frac{|\Phi_i(k)|^2}{1 + T_s \lambda_{\min}(\Gamma_i) |\Phi_i(k)|^2} \quad (\text{A.21})$$

which in turn implies that the normalized Φ_i vector signal is bounded, i.e., $\frac{|\Phi_i|}{m_i} \in l_\infty$. Therefore, the normalized estimation error ϵ_i is also bounded, i.e., $\epsilon_i \in l_\infty$, since it can be written as follows:

$$\epsilon_i(k) = \frac{-\Theta_i^{*T} \Phi_i(k)}{m_i^2(k)} \quad (\text{A.22})$$

Using Eq. (A.22) we can derive that the quantity $\epsilon_i m_i \in l_\infty$ is also bounded, since we can write:

$$\epsilon_i(k) m_i(k) = \frac{-\Theta_i^{*T} \Phi_i(k)}{m_i(k)} = \frac{-\tilde{\Theta}_i^T(k-1) \Phi_i(k)}{m_i(k)} \quad (\text{A.23})$$

where $\tilde{\Theta}_i(k) = \Theta_i(k) - \Theta_i^*$ is the estimation error of the gain.

The projection operator in Eq. (A.19) confines the estimate Θ_i inside the convex set S_i , which is a hyperbox that is created by all the lower and upper bound constraints $[L_i]_s$, $[U_i]_s$. Then, Eq. (A.19) can be rewritten as follows:

$$\Theta_i(k) = \Theta_i(k-1) + T_s \Gamma_i \epsilon_i(k) \Phi_i(k) + g(k) \quad (\text{A.24})$$

where

$$g(k) = \begin{cases} 0 & \text{if } \Theta_i(k-1) + T_s \Gamma_i \epsilon_i(k) \Phi_i(k) \in S_i \\ \tilde{\Theta}_i(k) - \tilde{\Theta}_i(k) & \text{if } \Theta_i(k-1) + T_s \Gamma_i \epsilon_i(k) \Phi_i(k) \notin S_i \end{cases} \quad (\text{A.25})$$

where $\tilde{\Theta}_i(k) = \Theta_i(k-1) + T_s \Gamma_i \epsilon_i(k) \Phi_i(k)$ and $\hat{\Theta}_i(k)$ is the orthogonal projection of $\tilde{\Theta}_i(k)$ to S_i . The projection operator guarantees that $\Theta_i \in l_\infty$.

Next, we consider the following Lyapunov-like function:

$$V_i(k) = \tilde{\Theta}_i^T(k) \Gamma_i^{-1} \tilde{\Theta}_i(k) \quad (\text{A.26})$$

Then, using Eq. (A.26), we define the following difference:

$$\begin{aligned} V_i(k) - V_i(k-1) &= \tilde{\Theta}_i^\top(k-1)\Gamma_i^{-1}\tilde{\Theta}_i(k-1) + T_s^2\epsilon_i^2(k)\Phi_i^\top(k)\Gamma_i\Phi_i(k) \\ &\quad + g_i^\top(k)\Gamma_i^{-1}g_i(k) + 2\tilde{\Theta}_i^\top(k-1)T_s\epsilon_i(k)\Phi_i(k) \\ &\quad + 2g_i^\top(k)T_s\epsilon_i(k)\Phi_i(k) + 2g_i^\top(k)\tilde{\Theta}_i(k-1) \\ &\quad - \tilde{\Theta}_i^\top(k-1)\Gamma_i^{-1}\tilde{\Theta}_i(k-1) \end{aligned} \quad (\text{A.27})$$

The following three terms from (A.27) satisfy:

$$\begin{aligned} &g_i^\top(k)\Gamma_i^{-1}g_i(k) + 2g_i^\top(k)(\tilde{\Theta}_i(k-1) + T_s\epsilon_i(k)\Phi_i(k)) \\ &\leq \lambda_{\max}(\Gamma_i^{-1})(\tilde{\Theta}_i(k) - \tilde{\Theta}_i(k))^\top(\tilde{\Theta}_i(k) - \tilde{\Theta}_i(k)) \\ &\quad + 2(\tilde{\Theta}_i(k) - \tilde{\Theta}_i(k))^\top(\tilde{\Theta}_i(k) - \Theta_i^*(k)) \\ &\leq c(-2(\tilde{\Theta}_i(k) - \Theta_i^*(k))^\top(\tilde{\Theta}_i(k) - \tilde{\Theta}_i(k)) \\ &\quad - |\tilde{\Theta}_i(k) - \tilde{\Theta}_i(k)|^2) \leq 0 \end{aligned} \quad (\text{A.28})$$

where the last inequality is obtained by using $(\tilde{\Theta}_i - \Theta_i^*)^\top(\tilde{\Theta}_i - \tilde{\Theta}_i) \geq 0$ because of the projection property. Using (A.28) in (A.27) we obtain:

$$\begin{aligned} V_i(k) - V_i(k-1) &= T_s^2\epsilon_i^2(k)\Phi_i^\top(k)\Gamma_i\Phi_i(k) + 2\tilde{\Theta}_i^\top(k-1)T_s\epsilon_i(k)\Phi_i(k) \\ &= T_s^2\epsilon_i^2(k)\Phi_i^\top(k)\Gamma_i\Phi_i(k) - 2T_s\epsilon_i^2(k)m_i^2(k) \\ &= -2T_s\epsilon_i^2(k)m_i^2(k)\left(1 - \frac{T_s\Phi_i^\top(k)\Gamma_i\Phi_i(k)}{2m_i^2(k)}\right) \end{aligned} \quad (\text{A.29})$$

Since for the chosen m_i we have that $\frac{T_s\Phi_i^\top(k)\Gamma_i\Phi_i(k)}{m_i^2(k)} < 1$, it follows that:

$$V_i(k) - V_i(k-1) \leq -T_s\epsilon_i^2(k)m_i^2(k) \leq 0 \quad (\text{A.30})$$

Using the last inequality, we derive the following one:

$$\sum_{k=1}^M \epsilon_i^2(k)m_i^2(k) \leq c(V_i(0) - V_i(M)) \quad (\text{A.31})$$

for some positive finite constant $c > 0$ and any positive integer $M > 0$. In addition, since the gain estimate is bounded, i.e. $\Theta_i \in l_\infty$, it follows that the Lyapunov-like function Eqn A.26 is bounded too, i.e. $V_i \in l_\infty$. Therefore, it follows from (A.31) that $\epsilon_i(k)m_i(k) \in l_2$ which implies that $\epsilon_i(k)m_i(k) \rightarrow 0$ as $k \rightarrow \infty$. Finally, because of the adaptive law (A.19), we have the following relation:

$$\Theta_i(k) - \Theta_i(k-1) \leq T_s\Gamma_i\epsilon_i(k)m_i(k)\frac{\Phi_i(k)}{m_i(k)} \quad (\text{A.32})$$

which implies that $|\Theta_i(k) - \Theta_i(k-1)| \in l_2$ and, thus, $|\Theta_i(k) - \Theta_i(k-1)| \rightarrow 0$ as $k \rightarrow \infty$.

To summarize, the estimation scheme (16) has the following properties:

- (i) $\Theta_i(k), \epsilon_i(k), \epsilon_i(k)m_i(k) \in l_\infty$
- (ii) $\epsilon_i(k)m_i(k), |\Theta_i(k) - \Theta_i(k-1)| \in l_2$
- (iii) $\epsilon_i(k)m_i(k), |\Theta_i(k) - \Theta_i(k-1)| \rightarrow 0$, as $k \rightarrow \infty$

In order to prove that temperature tracking is guaranteed even without prior knowledge of system parameters, we define the following signal:

$$m_f^2(k) \triangleq 1 + \sum_{i=1}^N \|e_{i,k-1}\|_{2\delta}^2 \quad (\text{A.33})$$

where $\|\cdot\|_{2\delta}$ is the $l_{2\delta}$ norm, as defined in [54], for some $\delta \in (A_d^{-2}, 1] \forall i$, so that $\frac{1}{\sqrt{\delta z + A_d}}$ has stable poles $\forall i$, where $A_d =$

$\max_{\forall i, \forall j \in N_i} (a_{m_i}, a_{w_{i,j}})$ and $a_{w_{i,j}} = e^{-2\frac{U_{w_{i,j}} A_{w_{i,j}} T_s}{C_{w_{i,j}}}}$. We can rewrite the zone temperature tracking error dynamics as follows:

$$e_i = \frac{z}{z - a_{m_i}} [\epsilon_i m_i^2] \quad (\text{A.34})$$

In addition, from the definition of m_i , we derive the following inequality:

$$m_i^2(k) \leq 1 + T_s \lambda_{\max}(\Gamma) |\Phi_i(k)|^2 \quad (\text{A.35})$$

and for the elements of Φ_i , we have that for some finite $c > 0$, $|T_{w_{i,j}}(k-1)| \leq c\|(T_{z_i})_{k-1}\|_{2\delta} + c\|(T_{z_p})_{k-1}\|_{2\delta} + c$ from (1b), (1c), $|T_{z_i}(k-1)| \leq |e_i(k-1)| + c$, $T_{m_i}, T_o \in l_\infty$ and, thus, $|T_{s a_i}(k-1)| \leq c|e_i(k-1)| + c\sum_{p \in N_i} \|e_{p,k-1}\|_{2\delta} + c\sum_{p \in N_i} |e_p(k-1)| + c$, using Lemma A.12.33.ii of [54]. In addition, using Lemma A.12.33.ii of [54] and considering $H(z) = 1$, we have that $|e_i(k)| \leq \|e_{i,k}\|_{2\delta}$. Therefore, $|\Phi_i(k)| \leq c\sum_{i=1}^N \|e_{i,k-1}\|_{2\delta}^2 + c$. That means that the signal m_i is bounded by the signal m_f :

$$m_i \leq c m_f \quad (\text{A.36})$$

for some finite $c > 0$.

Applying Lemma A.12.33 from [54] and since $\frac{z}{z - a_{m_i}}$ is analytic in $|z| \geq \sqrt{\delta}$, Eq. (A.34) gives the following inequality for some finite $c > 0$:

$$\|e_{i,k}\|_{2\delta} \leq c\|\epsilon_i m_i^2\|_{2\delta} \quad (\text{A.37})$$

which, combined with inequality (A.36), produces the following inequality:

$$\|e_{i,k}\|_{2\delta} \leq c\|\epsilon_i m_i m_{f,k}\|_{2\delta} \quad (\text{A.38})$$

Combining relations (A.33) and (A.38), we derive the following inequality:

$$m_f^2(k) \leq 1 + c\|\tilde{g}_{k-1} m_{f,k-1}\|_{2\delta}^2 \quad (\text{A.39})$$

where

$$\tilde{g}^2 = \sum_{i=1}^N |\epsilon_i m_i|^2 \quad (\text{A.40})$$

By applying Lemma A.12.31 from [54] to inequality (A.39) and using the fact that the geometric mean of a series is less than the arithmetic mean, for some finite $c > 0$ we obtain the following inequality:

$$\begin{aligned} m_f^2(k) &\leq 1 + c \sum_{p=0}^{k-1} \delta^{k-p} \tilde{g}^2(p) m_f^2(p) \\ &\leq 1 + c \sum_{p=0}^{k-1} \left[\delta^{k-p} \tilde{g}^2(p) \prod_{p < q < k} (1 + \delta^{k-q} \tilde{g}^2(q)) \right] \\ &\leq 1 + c \sum_{p=0}^{k-1} \left[\delta^{k-p} \tilde{g}^2(p) \left(1 + \frac{1}{k-p-1} \right. \right. \\ &\quad \left. \left. \times \sum_{q=p+1}^{k-1} (\delta^{k-q} \tilde{g}^2(q)) \right)^{k-p-1} \right] \end{aligned} \quad (\text{A.41})$$

From the structure of $\tilde{g}$ it follows that $\tilde{g} \in l_2$. In addition, since $\delta < 1$ we derive the following relation:

$$\begin{aligned} m_f^2(k) &\leq 1 + c \sum_{p=0}^{k-1} \left[\delta^{k-p} \tilde{g}^2(p) \left(1 + \frac{c}{k-p-1} \right)^{k-p-1} \right] \\ &\leq 1 + c \sum_{p=0}^{k-1} [\delta^{k-p} \tilde{g}^2(p) e^c] \end{aligned} \quad (\text{A.42})$$

for some finite $c > 0$. Therefore, again since $\tilde{g} \in l_2$, for $\delta < 1$, we conclude that the signal m_f is bounded, i.e., $m_f \in l_\infty$, since the series in the last inequality converges. Hence, all temperature tracking errors $e_i \in l_\infty \forall i$ are bounded too, due to the definition of m_f (A.33). Hence, signals $\Phi_i, m_i \in l_\infty$ are also bounded.

In addition, from (16c) we have that $z_i = \epsilon_i m_i^2 = \epsilon_i m_i m_i$. Since $\epsilon_i m_i \in l_2$ and $m_i \in l_\infty$, then $z_i \in l_2$. This means that $\Theta_i^{*T} \Phi_i \in l_2$ and $\Theta_i^{*T} \Phi_i \rightarrow 0$. Thus, from:

$$e_i = \frac{z}{z - a_{m_i}} [\Theta_i^{*T} \Phi_i] \quad (\text{A.43})$$

we have that for all zones the temperature tracking error is in l_2 . This implies that $\forall i \ e_i \rightarrow 0$ as $k \rightarrow \infty$. $\square$

CRediT authorship contribution statement

Georgios Lympereopoulos: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, Visualization. **Petros Ioannou:** Conceptualization, Methodology, Writing - review & editing, Supervision.

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